

University of Heidelberg
Faculty of Physics and Astronomy
Beginners Physics Lab 1

Experiment 13 – Resonance

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AI-Disclaimer

AI tools were used to assist in translating, as well as in the generation of text in section 1 as well as Fundamentals ??

1. Introduction

In this experiment, the fundamental dynamic properties of oscillating systems are investigated using a torsion pendulum. The focus is on both the free, damped oscillation and the forced oscillation under the influence of a periodic external torque.

For the experimental setup, the torsion pendulum is driven by a stepper motor with an eccentric, while an eddy current brake (based on Lenz's law) provides a controllable damping that is proportional to the velocity. The aim of the experiment is to determine the oscillation period and the damping constants of the system. Furthermore, the resonance behavior during forced oscillation is analyzed to experimentally determine characteristic quantities such as the resonance frequency, the full width at half maximum (FWHM) of the resonance curve, and the resonance amplitude enhancement, and to compare them with theoretical expectations.

2. Fundamentals

2.1. Free Damped Oscillation

If the torsion pendulum is displaced once and left to itself, it performs a free oscillation. Since frictional forces are present in every real system—which in this experiment (due to the eddy current brake) are proportional to the instantaneous velocity—the amplitude decreases as time progresses. The time dependence of this damped oscillation is described by the following equation:

$$a(t) = a_0 e^{-\delta t} \sin(\omega_f t) \quad (2.1)$$

Here, a_0 denotes the initial amplitude, δ the damping constant, and ω_f the angular frequency of the damped oscillation. The term $e^{-\delta t}$ describes the exponential decay of the oscillation. If one considers only the turning points of the oscillation (where the sine takes the value of one), the temporal decrease of the amplitude is given by:

$$a(t) = a_0 e^{-\delta t} \quad (2.2)$$

The damping constant δ can be determined experimentally by measuring the time $t_{1/2}$ after which the amplitude has dropped to exactly half of its original value. From equation (2) it follows:

$$\delta = \frac{\ln(2)}{t_{1/2}} \quad (2.3)$$

Due to damping, the system oscillates slightly slower than in the completely undamped case. The relationship between the angular frequency of the damped oscillation ω_f and the natural angular frequency of the undamped oscillation ω_0 is:

$$\omega_f^2 = \omega_0^2 - \delta^2 \quad (2.4)$$

2.2. Forced Oscillation and Resonance

If a periodic torque with the driving frequency ω is imposed on the system via a mechanical transmission system, it is referred to as a forced oscillation. After the initial transient process has decayed, the pendulum enters a stationary oscillation whose frequency exactly matches the driving frequency ω .

The amplitude $b(\omega)$ of this forced oscillation strongly depends on the applied frequency and is described by the resonance curve:

$$b(\omega) = \frac{A\omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\delta\omega)^2}} \quad (2.5)$$

Here, A represents the constant amplitude of the exciter (where for the static case $\omega \rightarrow 0$, the approximation $\omega_0 \approx \omega_f$ can be assumed). By differentiating this function, the resonance frequency ω' at which the pendulum's amplitude reaches its maximum can be determined:

$$\omega'^2 = \omega_0^2 - 2\delta^2 \quad (2.6)$$

As can be seen from equation (6), due to the damping, the amplitude maximum ω' occurs at a slightly lower frequency than the undamped natural frequency ω_0 .

Two other central quantities characterize the shape of the resonance curve:

- **Full Width at Half Maximum (FWHM) H :** It describes the frequency width of the resonance curve at the height $b(\omega')/\sqrt{2}$. For systems that are not too strongly damped, the following approximation applies:

$$H = 2\delta \quad (2.7)$$

- **Resonance Step-Up (Quality Factor):** It is defined as the ratio of the maximum amplitude at resonance to the static amplitude at a frequency of zero ($\omega \rightarrow 0$):

$$Q = \frac{b(\omega')}{b(\omega \rightarrow 0)} = \frac{\omega_0}{2\delta} \quad (2.8)$$

2.3. Transient Process

The complete mathematical solution of the equation of motion for a harmonic excitation results from the superposition of the free, decaying oscillation and the stationary, forced oscillation:

$$\alpha(t) = a_0 \sin(\omega_f t - \beta) e^{-\delta t} + b(\omega) \sin(\omega t - \epsilon) \quad (2.9)$$

The first term of this equation describes the transient process. The parameters β (depending on the initial state) and ϵ (the phase difference between the exciter and the system) determine the exact phase. As soon as the damped component has practically decayed to zero ($e^{-\delta t} \approx 0$), only the stationary oscillation remains, which is relevant for measuring the resonance curve.

3. Lab report

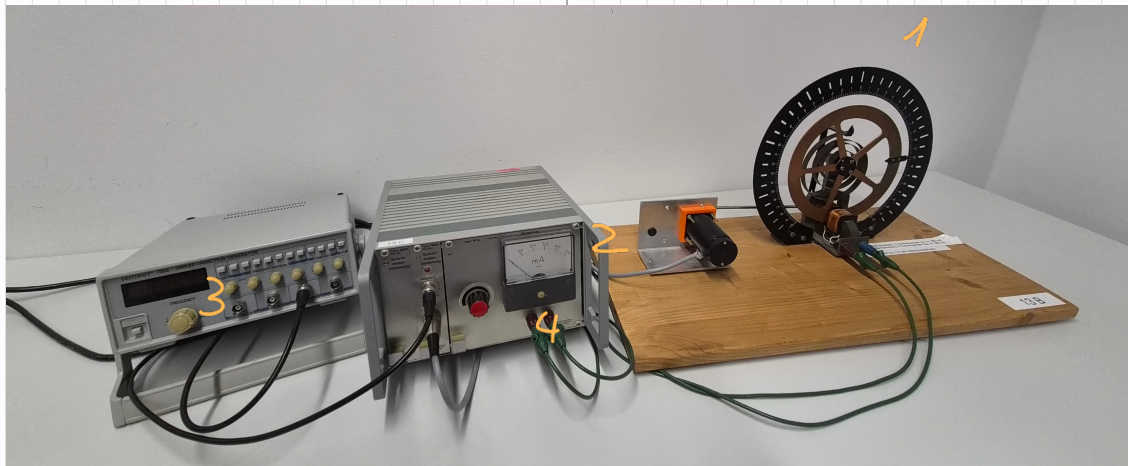
3.1. Setup and Exercises

23.09.2026

Theo Senoner & Jonathan Gießen

V13: Resonance

Tutor: Wiktor Poklepa

T1:Fig 3.1: Setup

- 1 • Rotary pendulum (Pohl's wheel), stimulated by a stepper motor with an eccentric
- 2 • Stepper motor drive controller with power supply
- 3 • Function generator
- 4 • Power supply unit for the eddy current brake

T2: $\Delta t = \pm 0,25s$ Tab 3.2: Oscillation period

Messung	n	t [s]
1	20	39,61
2	20	39,72
3	20	39,73

T3: $\Delta t = \pm 0,5s$

Tab 3.3 Transient time

Current [mA]	Oscillations until 5% Amplitude [n]	Time [s]
$I_1: 280$	14	27,7
$I_2: 360$	9	17,8
$I_1: 280$	14	26,0
$I_2: 360$	9	17,7

Δt is higher because reading off the amplitude comes with uncertainty, too

$\Delta I = \pm 12,5 \text{ mA}$ (b. the scale)

T4: $\Delta L = 0.2 \text{ cm}$ Tab. 3.4: Amplitude decrease (damped)

n	Ampl. [cm] I_1	Ampl. [cm] I_2	Ampl. [cm] I_1	Ampl. [cm] I_2
0	20,0	20,0	20,0	20,0
1	16,5	16,2	14,6	14,8
2	13,6	13,5	10,7	10,7
3	11,2	11,1	7,8	7,8
4	9,2	9,0	5,6	5,6
5	7,6	7,4	4,1	4,1
6	6,2	6,0	2,9	2,9
7	5,0	5,0	2,0	2,0
8	4,0	4,0	1,4	1,4
9	3,2	3,2	1,0	1,0
10	2,6	2,6	0,7	0,6
11	2,1	2,0	0,5	0,2
12	1,7	1,6	0,3	
13	1,3	1,3		
14	1,0	1,0		
15	0,8	0,8		
<u>16</u>	0,6	0,6		
<u>17</u>	0,5	0,4		

T5:

 $\Delta l = \pm 0,2 \text{ cm}$ Tab 3.5: Amplitude decrease (forced) I_1

$f [\text{Hz}]$	Ampl. $b [\text{cm}] (I_1)$	$f [\text{Hz}]$	Ampl. $b [\text{cm}] (I_1)$
300	0,4	2150	2,6
500	0,4	2200	2,0
700	0,4	2250	1,6
900	0,5	2300	1,3
1100	0,55	2500	0,8
1300	0,7	2700	0,6
1500	0,9	2900	0,4
1600	1,0	3100	0,3
1700	1,3		
1800	1,8		
1850	2,2		
1900	2,8		
1950	3,8		
1960	4,3		
1970	4,6		
1980	4,9		
1990	5,3		
2000	5,7		
2010	5,9		
2020	6,1		
2030	6,1		
2040	6,0		
2050	5,7		
2060	5,2		
2070	4,8		
2080	4,5		
2090	4,2		
2100	3,8		

Tab 3.6: Amplitude decrease (forced) I_2

f [Hz]	Ampl. b [cm] (I_2)	f [Hz]	Ampl. b [cm] (I_2)
300	0,4	2150	2,3
500	0,5	2200	1,8
700	0,5	2250	1,5
900	0,55	2300	1,3
1100	0,6	2500	0,7
1300	0,7	2700	0,6
1500	0,8	2900	0,4
1600	1,0	3100	0,4
1700	1,3		
1800	1,7		
1850	2,0		
1900	2,5		
1950	3,1		
1960	3,2		
1970	3,3		
1980	3,4		
1990	3,6		
2000	3,7		
2010	3,8		
2020	3,8		
2030	3,7		
2040	3,7		
2050	3,6		
2060	3,5		
2070	3,4		
2080	3,2		
2090	3,0		
2100	2,9		

2. RWM

3.2. Procedure

The measuring process is divided into four main tasks to analyze both free and forced oscillations:

1. **Period of the Free Oscillator:** Determine the oscillation period T_0 of the undamped, free torsion pendulum by measuring the duration of 20 full oscillation periods. Repeat this measurement three times.
2. **Transient Time and Damping:** Activate the eddy current brake. Find two specific damping currents: one where the amplitude decays to 5% of its initial value after about 10 oscillations, and another where it decays to 5% after about 15 oscillations. Measure the transient time for both settings.
3. **Amplitude Decay:** For the two selected damping currents, release the pendulum from a reversal point at $t = 0$ and record the amplitude after every full period to document the exponential decay.
4. **Forced Oscillations:** Excite the pendulum with the stepper motor. Measure the stationary amplitude as a function of the excitation frequency (generator frequency from 300 Hz to 4000 Hz) for both damping settings. Wait for the transient process to finish before each reading, and use smaller frequency steps (50 Hz) near the resonance peak to accurately determine the resonance frequency and the Full Width at Half Maximum (FWHM).

4. Examination of the Results

4.1. Excercise 1: Period of the Free Oscillator

We measured the oscillation period of the undamped torsion pendulum by timing 20 full oscillations. The measurements were repeated three times to ensure accuracy. The results are summarized in Table 4.1. The measured Values are:

Table 4.1: Measured oscillation periods for 20 full oscillations of the undamped torsion pendulum.

Measurement	Time for 20 Oscillations (s)	Calculated Period T_0 (s)
1	39.61 ± 0.25	1.981 ± 0.013
2	39.72 ± 0.25	1.986 ± 0.013
3	39.73 ± 0.25	1.987 ± 0.013

With the error calculation of:

$$\Delta T_0 = \sqrt{\Delta T_{\text{stat}}^2 + \Delta T_{\text{sys}}^2} = \sqrt{(0.002)^2 + (0.013)^2} \approx 0.014 \text{ s} \quad (4.1)$$

The Mean of the measured periods is:

$$T_0 = 1.98413 \quad (4.2)$$

Result

The oscillation period of the undamped torsion pendulum is determined to be:

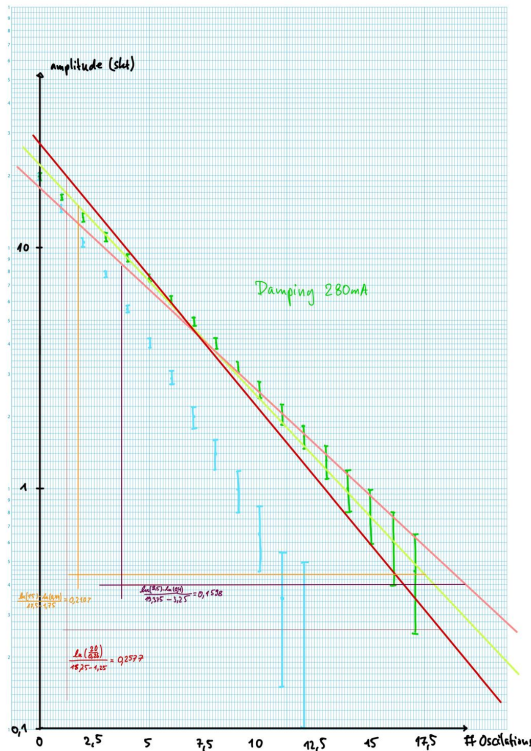
$$T_0 = (1.984 \pm 0.014) \text{ s} \quad (4.3)$$

$$f_0 = (0.504 \pm 0.004) \text{ Hz} \quad (4.4)$$

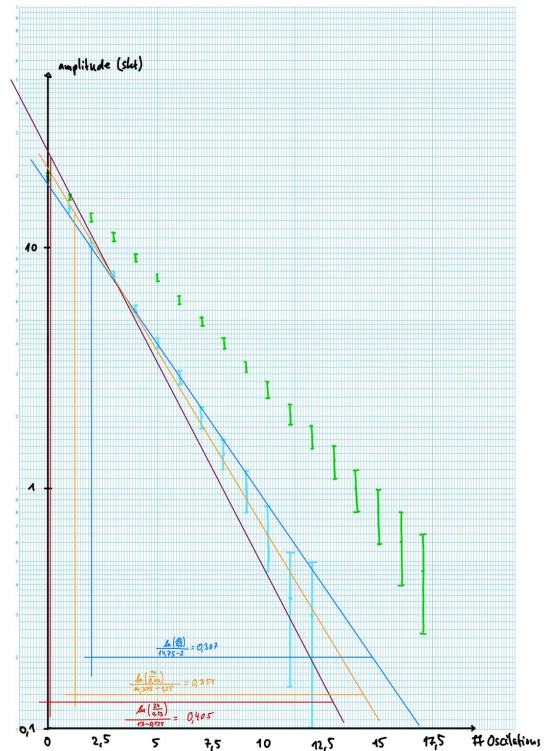
4.2. Excercise 2: Evaluate the damping constant δ

The damping constant δ can be determined by measuring the amplitude with respect to the number of oscillations. For each damping current of the eddy current brake, the average of two measurements of the amplitude was determined after each full oscillation until it reached $> 5\%$ of the initial amplitude. The results are plotted logarithmically to determine the damping constant δ from the slope of the fit.

The error of the amplitude is estimated to be $\Delta a = 0.2 \text{ Skt}$, which corresponds to the uncertainty of the measurement due to the scale division of the torsion pendulum.



(a) Damping current: 280 mA.



(b) Damping current: 360 mA.

Figure 4.1: Determination of the damping coefficient for two damping currents.

from the slope of the linear fit, the damping constants are:

1. for 280 mA: $(-0.21 \pm 0.05) \frac{1}{\text{Oscillations}}$

¹The plots are listed in full scale in the appendix

2. for 360 mA: $(-0.35 \pm 0.05) \frac{1}{\text{Oscillations}}$

Since the oscillation period T_0 is known, the damping constants can be converted to units of $\frac{1}{s}$:

$$\delta_{280} = \frac{-0.21(5)}{T_0} = (-0.11 \pm 0.03) \frac{1}{s} \quad (4.5)$$

$$\delta_{360} = \frac{-0.35(5)}{T_0} = (-0.18 \pm 0.03) \frac{1}{s} \quad (4.6)$$

Result

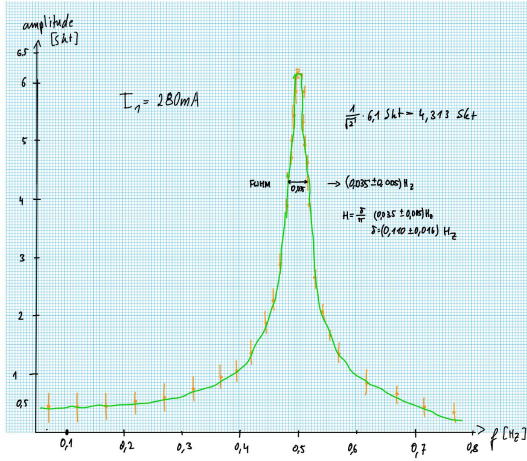
The damping constants for the two damping currents are determined to be:

$$\delta_{280} = (-0.11 \pm 0.03) \frac{1}{s} \quad (4.7)$$

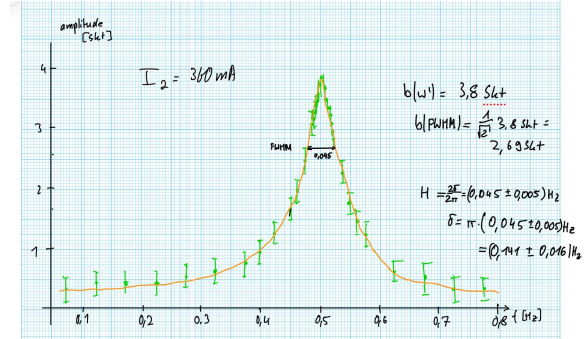
$$\delta_{360} = (-0.18 \pm 0.03) \frac{1}{s} \quad (4.8)$$

4.3. Excercise 3: Forced Oscillation and Resonance

The resonance curves are drawn for both damping currents.



(a) Damping current: 280 mA.



(b) Damping current: 360 mA.

Figure 4.2: Resonance curves for two damping currents. The resonance frequency and the Full Width at Half Maximum (FWHM) are determined from the plots.

The correlation between the damping constant δ and the FWHM H is given by equation 2.7, the measured width has to be multiplied by π to get the damping constant. The measured values for the resonance frequency and the Full Width at Half Maximum (FWHM) are summarized in Table 4.2.

Table 4.2: Measured resonance frequencies and Full Width at Half Maximum (FWHM) for two damping currents.

Damping Current	Resonance Frequency f (Hz)	δ (Hz)
280 mA	0.505 ± 0.005	0.110 ± 0.016
360 mA	0.506 ± 0.005	0.141 ± 0.016

The natural frequency $f_0 = (0.504 \pm 0.004)$ Hz is slightly lower than the measured resonance frequencies, which is consistent with the theoretical expectation that the resonance frequency is slightly lower than the natural frequency due to damping.

The resonance step-up (quality factor) (Equation 2.8) tells how much the amplitude is amplified at resonance compared to the static amplitude.

The quality factor can also be calculated using the measured values of the resonance frequency and the damping constant.

$$Q_{damping} = \frac{\omega_0}{2\delta} \quad (4.9)$$

$$Q_{measured} = \frac{b(\omega')}{b(\omega \rightarrow 0)} \quad (4.10)$$

Table 4.3: Quality factor for two damping currents.

Damping Current	$Q_{damping}$	$Q_{measured}$
280 mA	15 ± 4	15 ± 8
360 mA	9.0 ± 1.3	9 ± 5

The difference between the theoretical and measured damping constants is:

$$\sigma\delta = \frac{\delta_{measured} - \delta_{damping}}{\sqrt{\Delta\delta_{damping}^2 + \Delta\delta_{measured}^2}} = \begin{cases} \sigma\delta_{280} \approx 0.14 \\ \sigma\delta_{360} \approx 1.18 \end{cases} \quad (4.11)$$

Result

The measured resonance frequencies and Full Width at Half Maximum (FWHM) for the two damping currents are:

$$f_{280} = (0.505 \pm 0.005) \text{ Hz} \quad (4.12)$$

$$f_{360} = (0.506 \pm 0.005) \text{ Hz} \quad (4.13)$$

$$\delta_{280} = (0.110 \pm 0.016) \text{ Hz} \Rightarrow 0.14\sigma \quad (4.14)$$

$$\delta_{360} = (0.141 \pm 0.016) \text{ Hz} \Rightarrow 1.18\sigma \quad (4.15)$$

The quality factors for the two damping currents are:

$$Q_{280} = 15 \pm 4 \quad (4.16)$$

$$Q_{360} = 9.0 \pm 1.3 \quad (4.17)$$

5. Conclusion

In this experiment, we investigated the fundamental dynamic properties of a torsion pendulum, focusing on both free, damped oscillations and forced oscillations under periodic external torque. We determined the oscillation period of the undamped torsion pendulum, measured the damping constants for two different damping currents, and analyzed the resonance behavior during forced oscillations. The measured resonance frequencies and Full Width at Half Maximum (FWHM) were consistent with theoretical expectations, and the quality factors for the two damping currents were determined. With a sigma-deviation of 0.14 and 1.18. The bigger deviation can be explained by the approximation using the Lorentz profile, which is only exact in the limit of small damping. The damping with $I_2 = 360$ mA shows a larger deviation than the damping with $I_1 = 280$ mA. Using python, i've fitted the data to the exact Breit-Wigner function, which is the exact solution of the forced oscillation.

$$A(\omega) = \frac{A_0 \cdot \omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}} \quad (5.1)$$

using this fit, the damping constant is given by γ and the resonance frequency is given by ω_0 . The results are:

$$\gamma_{280} = 0.0981 \text{ Hz} \quad (5.2)$$

$$\gamma_{360} = 0.165 \text{ Hz} \quad (5.3)$$

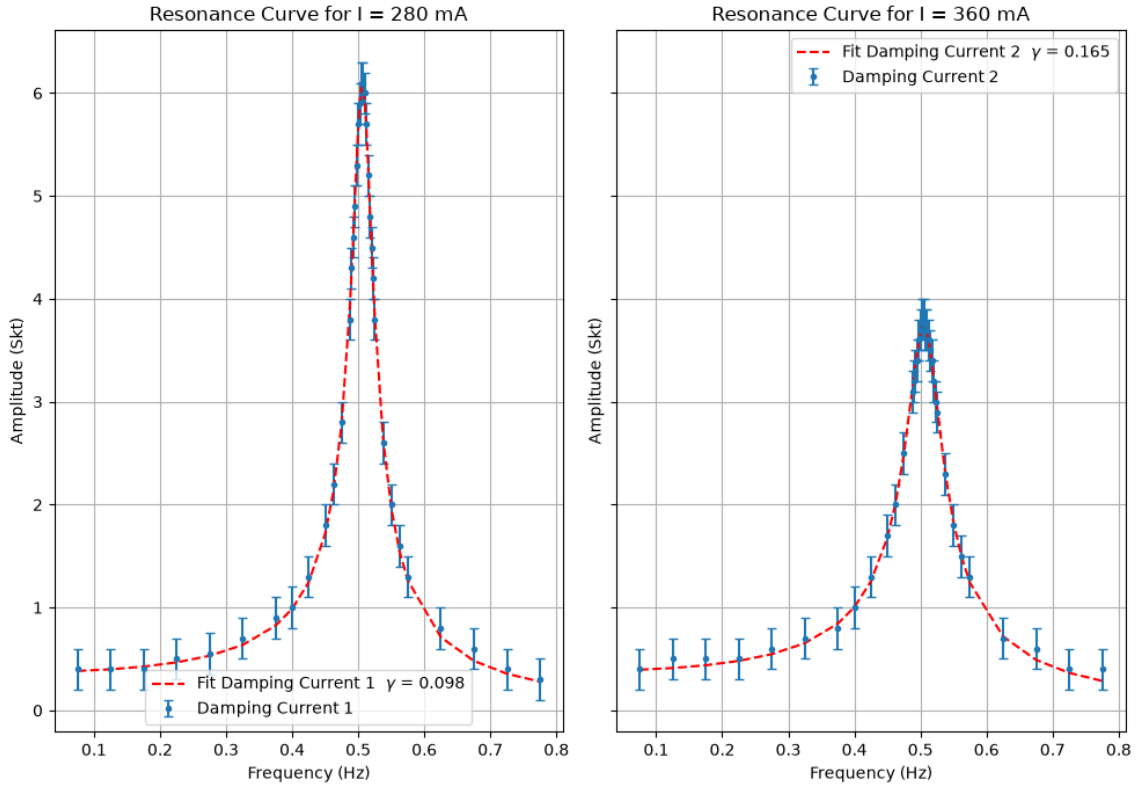


Figure 5.1: Resonance curve for a damping current of 280 mA and 360 mA fitted with the exact Breit-Wigner function.

This shows the exact deviation by assuming a Lorentz profile.

Overall, the experiment provided valuable insights into the behavior of oscillating systems and their response to external forces.

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References

- [1] Dr. J.Wagner, *Physikalisches Anfängerpraktikum*, Stand 11/2024
- [2] W. Ilberg, M. Krötzsch, *Physikalisches Praktikum für Anfänger*, BSB B. G. Teubner Verlagsgesellschaft, Leipzig (1985).

A. Plots

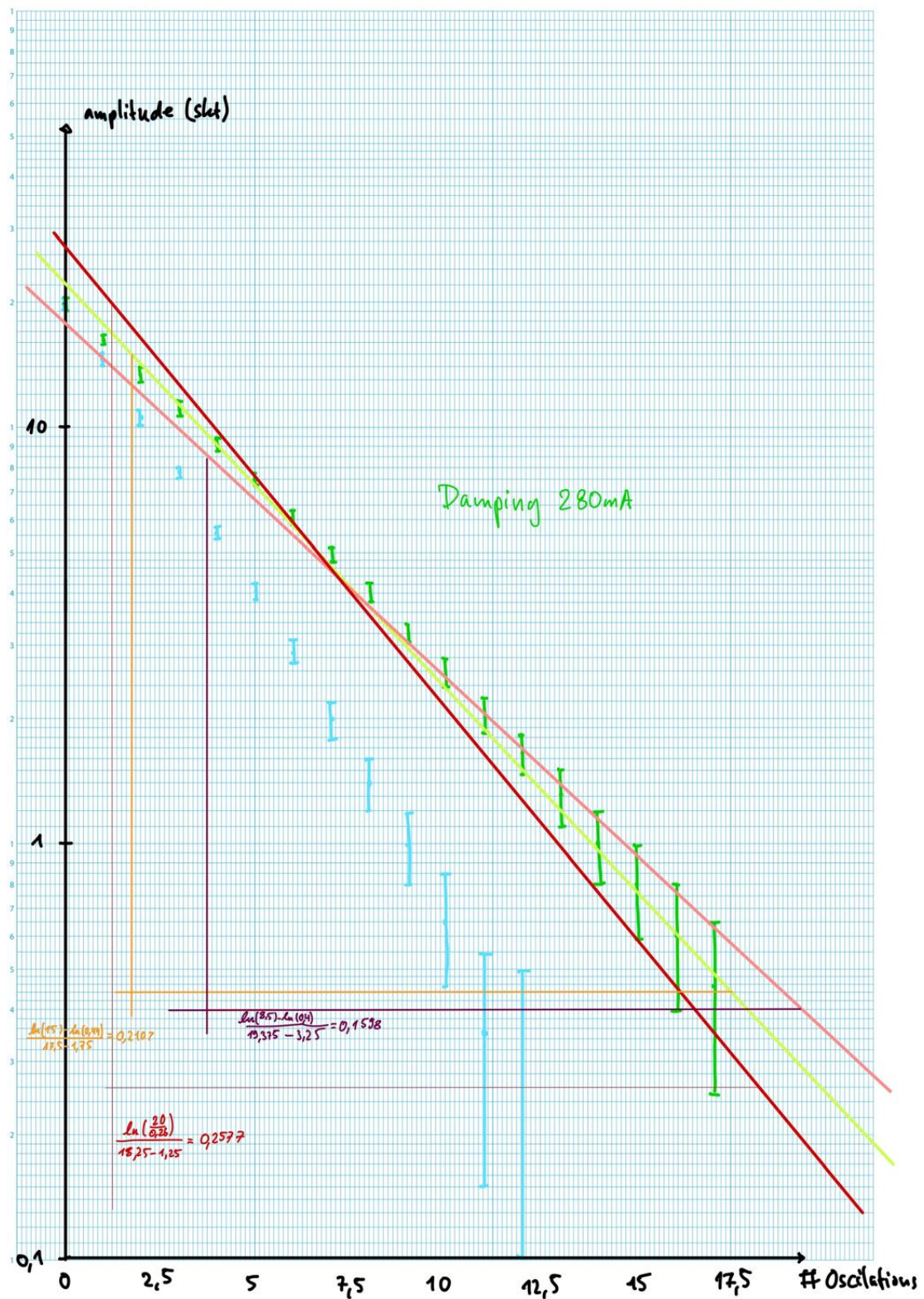


Figure A.1: Amplitude decay for a damping current of 280 mA.



Figure A.2: Amplitude decay for a damping current of 360 mA.

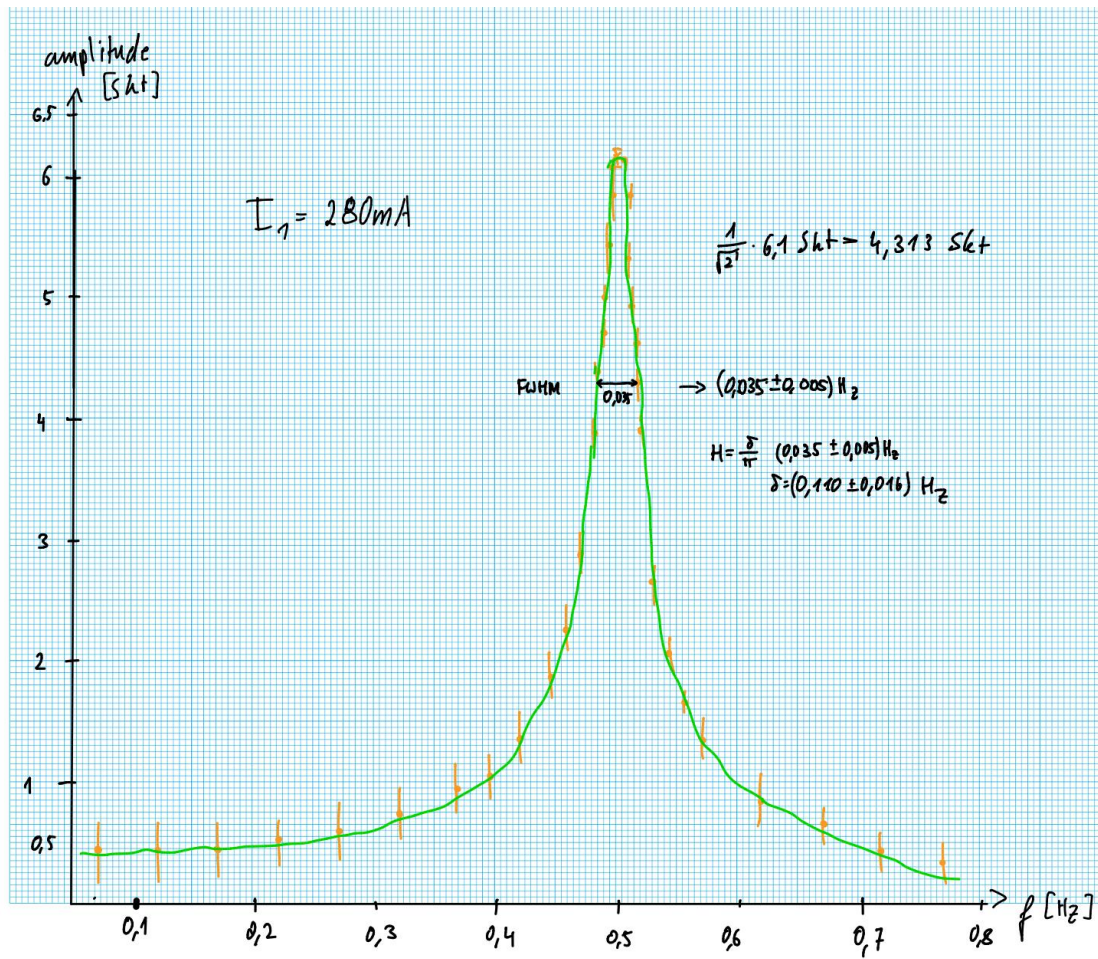


Figure A.3: Resonance curve for a damping current of 280 mA.

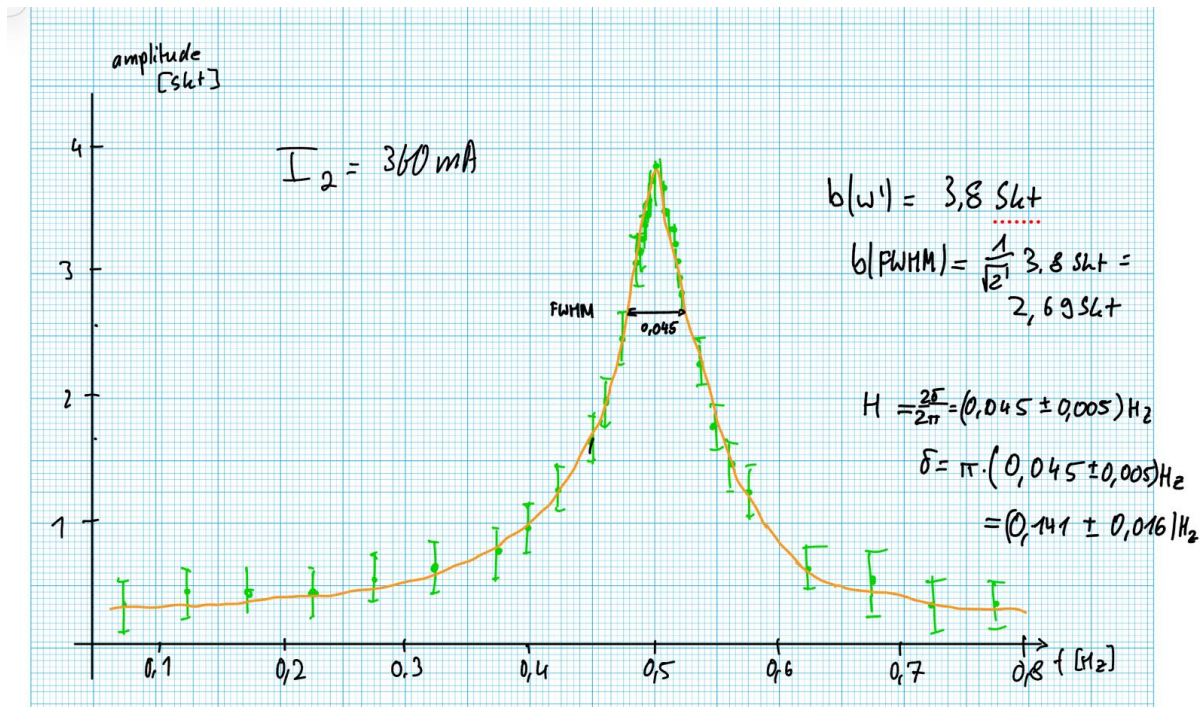


Figure A.4: Resonance curve for a damping current of 360 mA.

B. Python Code

The following code was used to analyze the data and generate the plots in this report. It includes the necessary functions for fitting the data to the theoretical models and calculating the relevant parameters.

T_0 berechnen

```
[1]: import numpy as np
import uncertainties as unc
import uncertainties.unumpy as unp
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit

[2]: T = np.array([39.61, 39.72, 39.73])

unc_T = 0.25

T = unp.uarray(T, unc_T)
T_0 = T/20
T_0
T_fin_err = np.sqrt((np.std(unp.nominal_values(T_0), ddof=1) / np.
    ↪sqrt(len(T_0)))*2 + 0.013**2)
T_fin = unc.ufloat(
    float(np.mean(unp.nominal_values(T_0))),
    float(T_fin_err)
)
print(T_fin)
print(1/T_fin)
```

1.984+/-0.013

0.5039+/-0.0033

```
[3]: i1_run1 = [20.0, 16.5, 13.6, 11.2, 9.2, 7.6, 6.2, 5.0, 4.0, 3.2, 2.6, 2.
    ↪1, 1.7, 1.3, 1.0, 0.8, 0.6, 0.5]
i1_run2 = [20.0, 16.2, 13.5, 11.1, 9.0, 7.4, 6.0, 5.0, 4.0, 3.2, 2.6, 2.
    ↪0, 1.6, 1.3, 1.0, 0.8, 0.6, 0.4]

i2_run1 = [20.0, 14.6, 10.7, 7.8, 5.6, 4.1, 2.9, 2.0, 1.4, 1.0, 0.7, 0.
    ↪5, 0.3]
i2_run2 = [20.0, 14.8, 10.7, 7.8, 5.6, 4.1, 2.9, 2.0, 1.4, 1.0, 0.6, 0.
    ↪2]

def calculate_averages(run1, run2, n_max):
    averages = []
    for i in range(n_max):
        values = []
        if i < len(run1): values.append(run1[i])
        if i < len(run2): values.append(run2[i])

        if values:
            avg = sum(values) / len(values)
            averages.append(avg)
    else:
```

```

        averages.append(None)
    return averages

# Process averages
i1_avg = calculate_averages(i1_run1, i1_run2, len(i1_run1))
i2_avg = calculate_averages(i2_run1, i2_run2, len(i2_run1))

#print("Averages for Damping Current 1:", i1_avg)
#print("Averages for Damping Current 2:", i2_avg)
#

```

```

[4]: plt.errorbar(range(len(i1_avg)), i1_avg, yerr=0.2, fmt='.', □
    ↪label='Damping Current 1', capsize=3)
plt.errorbar(range(len(i2_avg)), i2_avg, yerr=0.2, fmt='.', □
    ↪label='Damping Current 2', capsize=3)
plt.xlabel('Number of Oscillations')
plt.ylabel('Amplitude (Skt)')
plt.title('Amplitude vs Number of Oscillations')
plt.legend()
plt.yscale('log')
plt.grid()
plt.grid(which='minor', linestyle=':', linewidth=0.5)

popt1, pcov1 = curve_fit(lambda x, a, b: a * np.exp(b * x), □
    ↪range(len(i1_avg)), i1_avg, p0=(20, -0.2))
popt2, pcov2 = curve_fit(lambda x, a, b: a * np.exp(b * x), □
    ↪range(len(i2_avg)), i2_avg, p0=(20, -0.2))

print(f"Fitted parameters for Damping Current 1: a = {popt1[0]}, b = □
    ↪{popt1[1]}")
print(f"Fitted parameters for Damping Current 2: a = {popt2[0]}, b = □
    ↪{popt2[1]}")

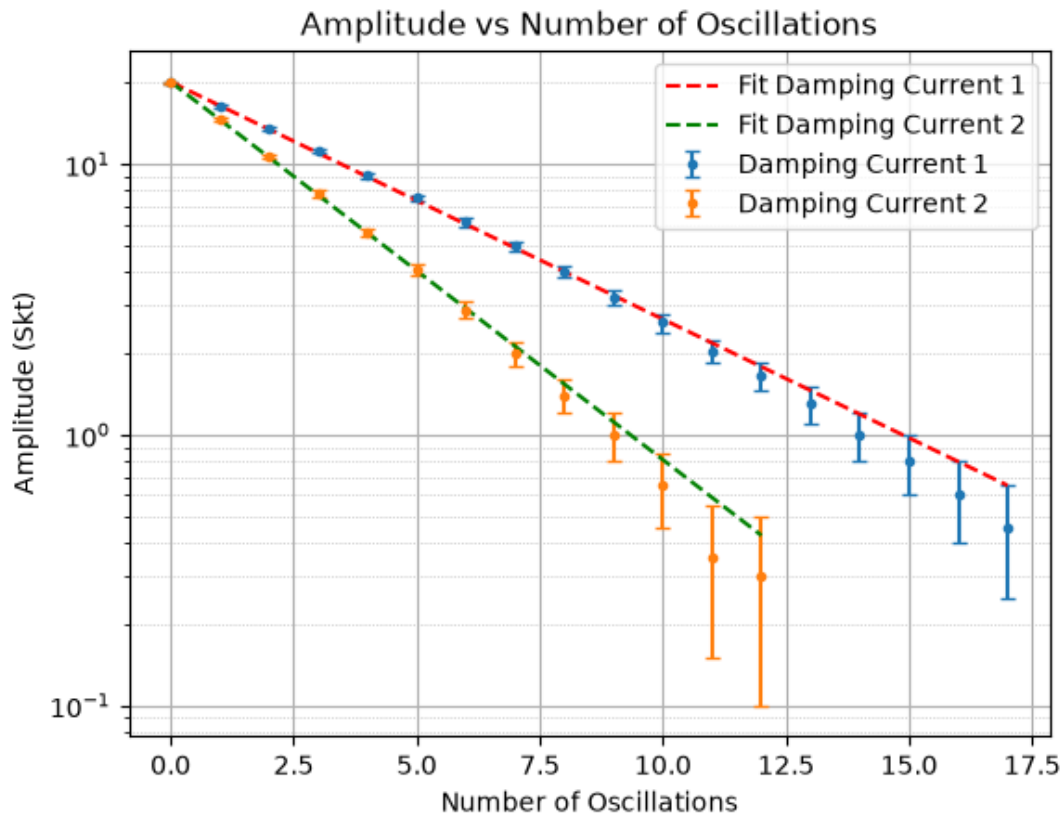
plt.plot(range(len(i1_avg)), popt1[0] * np.exp(popt1[1] * np.
    ↪array(range(len(i1_avg)))), 'r--', label='Fit Damping Current 1')
plt.plot(range(len(i2_avg)), popt2[0] * np.exp(popt2[1] * np.
    ↪array(range(len(i2_avg)))), 'g--', label='Fit Damping Current 2')
plt.legend()

```

Fitted parameters for Damping Current 1: a = 20.141418645517, b = -0.2019534759495409

Fitted parameters for Damping Current 2: a = 20.14544028297295, b = -0.3215491353551395

[4]: <matplotlib.legend.Legend at 0x1e5bf37c050>



```
[5]: d1 = unc.ufloat(-0.21, 0.05)
      d2 = unc.ufloat(-0.35, 0.05)

      delta_280 = d1/T_fin
      delta_360 = d2/T_fin
      print(f"Damping constant for 280 mA: {delta_280}")
      print(f"Damping constant for 360 mA: {delta_360}")
      print(f"Fitted by python: Damping constant for 280 mA (exponential):_
        ↪ {popt1[1]/T_fin}")
      print(f"Fitted by python: Damping constant for 360 mA (exponential):_
        ↪ {popt2[1]/T_fin}")
```

Damping constant for 280 mA: -0.106+/-0.025

Damping constant for 360 mA: -0.176+/-0.025

Fitted by python: Damping constant for 280 mA (exponential): -0.1018+/-0.
 ↪ 0007

Fitted by python: Damping constant for 360 mA (exponential): -0.1620+/-0.
 ↪ 0011

Excercise 3

[6]:


```

f = np.array([300, 500, 700, 900, 1100, 1300, 1500, 1600, 1700, 1800,
↳1850, 1900, 1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020, 2030,
↳2040, 2050, 2060, 2070, 2080, 2090, 2100, 2150, 2200, 2250, 2300,
↳2500, 2700, 2900, 3100]) / 4000

b_I1 = np.array([0.4, 0.4, 0.4, 0.5, 0.55, 0.7, 0.9, 1.0, 1.3, 1.8, 2.
↳2, 2.8, 3.8, 4.3, 4.6, 4.9, 5.3, 5.7, 5.9, 6.1, 6.1, 6.0, 5.7, 5.2,
↳4.8, 4.5, 4.2, 3.8, 2.6, 2.0, 1.6, 1.3, 0.8, 0.6, 0.4, 0.3])

b_I2 = np.array([0.4, 0.5, 0.5, 0.5, 0.6, 0.7, 0.8, 1.0, 1.3, 1.7, 2.0,
↳2.5, 3.1, 3.2, 3.3, 3.4, 3.6, 3.7, 3.8, 3.8, 3.7, 3.7, 3.6, 3.5, 3.
↳4, 3.2, 3.0, 2.9, 2.3, 1.8, 1.5, 1.3, 0.7, 0.6, 0.4, 0.4])

err = 0.2
omega = 2 * np.pi * f

def amplitude_curve(omega, A_0, omega_0, gamma):
    return A_0 * omega_0**2 / np.sqrt((omega_0**2 - omega**2)**2 + (2
↳* gamma * omega)**2)

fit_results = {}
for label, amplitudes in {"I1": b_I1, "I2": b_I2}.items():
    initial_guess = (0.2, omega[np.argmax(amplitudes)], 0.05)
    parameters, covariance = curve_fit(
        amplitude_curve,
        omega,
        amplitudes,
        p0=initial_guess,
        sigma=np.full_like(omega, err),
        absolute_sigma=True,
        bounds=([0, 0, 0], [np.inf, np.inf, np.inf]),
        x_scale="jac",
        maxfev=10000,
    )

    A_0, omega_0, gamma = parameters
    fit_results[label] = (parameters, covariance)
    print(f"{label}: A_0 = {A_0:.6f}, omega_0 = {omega_0:.6f}")
    print(f"{label}: gamma = {gamma:.6f}")
    print(f"{label}: gamma = {gamma * 4000:.3f} rad/s = {gamma * 4000 /
↳ (2 * np.pi):.3f} Hz")
    print(f"{label}: FWHM = {2 * gamma * 4000 / (2 * np.pi):.3f} Hz")

```

I1: A_0 = 0.374702, omega_0 = 3.183433

I1: gamma = 0.098050

I1: gamma = 392.200 rad/s = 62.421 Hz

```

I1: FWHM = 124.841 Hz
I2: A_0 = 0.386554, omega_0 = 3.182402
I2: gamma = 0.165132
I2: gamma = 660.529 rad/s = 105.126 Hz
I2: FWHM = 210.253 Hz

```

```

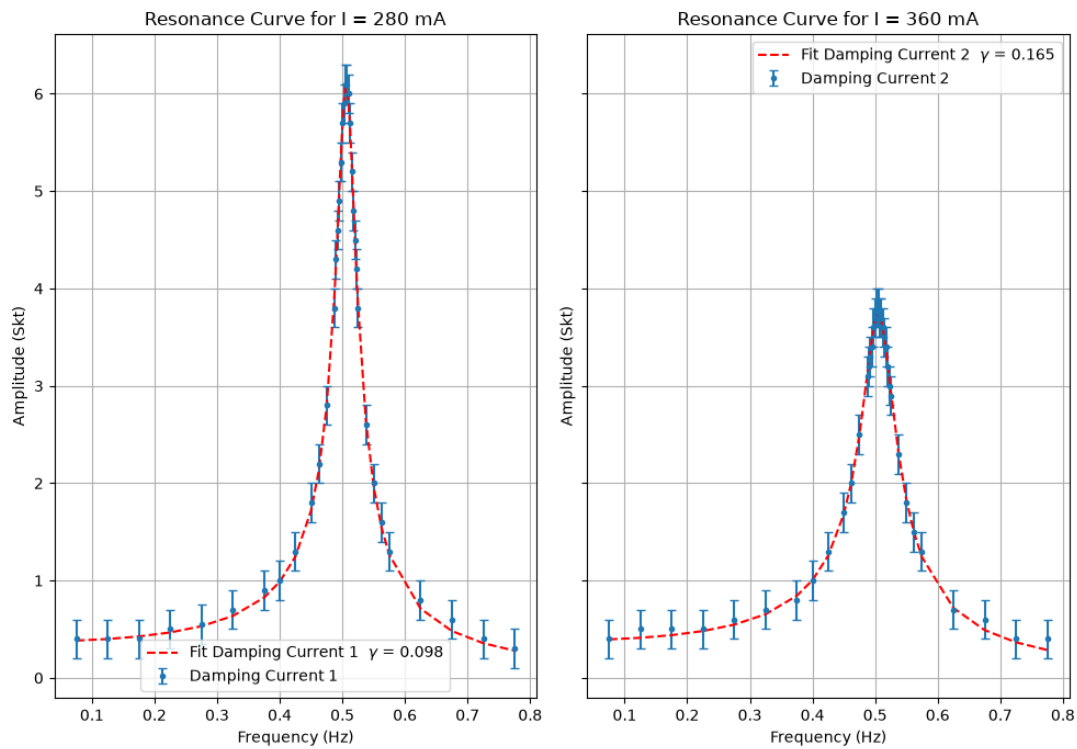
[22]: fig, (ax1, ax2) = plt.subplots(figsize=(10, 7), ncols=2, sharex=True,
    ↪sharey=True)

ax1.errorbar(f, b_I1, yerr=err, fmt='.', label='Damping Current 1',
    ↪capsize=3)
ax2.errorbar(f, b_I2, yerr=err, fmt='.', label='Damping Current 2',
    ↪capsize=3)
ax1.grid()
ax2.grid()

ax1.plot(
    f,
    amplitude_curve(2 * np.pi * f, *fit_results["I1"][0]),
    "r--",
    label=rf"Fit Damping Current 1  $\gamma$ = {fit_results['I1'][0][2]:
    ↪.3f}",
)
ax2.plot(
    f,
    amplitude_curve(2 * np.pi * f, *fit_results["I2"][0]),
    "r--",
    label=rf"Fit Damping Current 2  $\gamma$ = {fit_results['I2'][0][2]:
    ↪.3f}",
)

ax1.set_ylabel("Amplitude (Skt)")
ax2.set_ylabel("Amplitude (Skt)")
ax1.set_xlabel("Frequency (Hz)")
ax2.set_xlabel("Frequency (Hz)")
ax2.set_title("Resonance Curve for I = 360 mA")
ax1.set_title("Resonance Curve for I = 280 mA")
ax1.legend()
ax2.legend()
plt.tight_layout()

```



```
[8]: w_0_280 = unc.ufloat(0.505, 0.005) * 2 * 3.141
      w_0_360 = unc.ufloat(0.506, 0.005) * 2 * 3.141

      Q_280 = w_0_280 / (delta_280 * 2)
      Q_360 = w_0_360 / (delta_360 * 2)
      print(f"Quality factor for 280 mA: {Q_280:.3uP}")
      print(f"Quality factor for 360 mA: {Q_360:.3uP}")
```

Quality factor for 280 mA: -14.99 ± 3.57

Quality factor for 360 mA: -9.01 ± 1.29

```
[9]: win_280 = unc.ufloat(0.4, 0.2)
      win_360 = unc.ufloat(0.4, 0.2)

      wmax_280 = unc.ufloat(6.1, 0.2)
      wmax_360 = unc.ufloat(3.8, 0.2)

      print(f"Quality factor for 280 mA (measured): {wmax_280/win_280:.3uP}")
      print(f"Quality factor for 360 mA (measured): {wmax_360/win_360:.3uP}")
```

Quality factor for 280 mA (measured): 15.25 ± 7.64

Quality factor for 360 mA (measured): 9.50 ± 4.78

```
[10]: print(delta_280.s)
      sigma_err_delta_280 = (-delta_280.n - 0.110)/np.sqrt(delta_280.s**2 + 0.
      ↪016**2)
      sigma_err_delta_360 = (-delta_360.n - 0.141)/np.sqrt(delta_360.s**2 + 0.
      ↪016**2)

      print(f"Sigma error for delta_280: {sigma_err_delta_280:.3f}")
      print(f"Sigma error for delta_360: {sigma_err_delta_360:.3f}")
```

0.02520712463532635

Sigma error for delta_280: -0.140

Sigma error for delta_360: 1.184

```
[ ]:
```